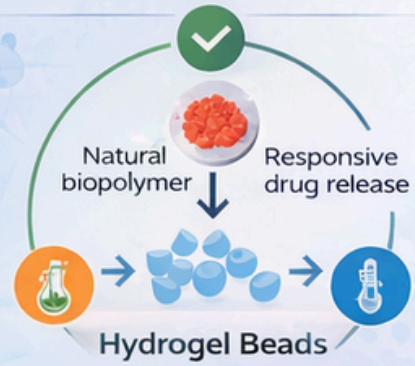
Smart Biopolymer (Alginate Hydrogel) for Temperature-Controlled Drug Release

Introduction

Objectives

- Develop temperature-responsive, drug-loaded alginate hydrogel.
- Demonstrate controlled release model drug (food dye) at different temperatures.

Smart Biomaterial Chart



Materials & Methods

Items/ Tools

- Sodium alginate powder
- Calcium chloride granules
- Food dye (model drug)
- Thermometer & gloves
- Dropper & syringe
- Boiled tap water



Hydrogels Preparation

Drug Model: Concentrate food color solution prepared to water Alginate Solution.

- Pouring 3-5 drops of Food Color solution in the Sodium Alginate Solution.
- Mix the solution until Homogenized.

Bead Formation:

- Use dropper/syringe to take Alginate Solution and drop into Calcium Chloride water Solution.
- Wait for 5 minutes, rinse with water.

Observation and Table

Temperature	Observation
25°C	Slow drug release
37°C	Fast drug release

Results



At Room Temperature (~25°C)

At Body-like Temperature (37°C)

Conclusion

- This suggest alginate-calcium hydrogel's potential as a smart biomaterial for controlled drug delivery.
- Fine-tune the hydrogel's temperature sensitivity
- Study more complex drug release profiles.
- Explore biomedical applications such as wound healing & tissue engineering

Group Members